

Private Lander Carrying NASA Experiments Lands on the Moon

A private spacecraft carrying several science experiments has successfully landed on the moon.

The lander, called Blue Ghost, was built by the American company Firefly Aerospace. The spacecraft touched down March 2 on the part of the moon's near side called Mare Crisium. The near side of the moon is the side facing Earth.

Officials at Firefly's Mission Control center near Austin, Texas, confirmed the successful landing. The chief engineer for the Blue Ghost mission, Will Coogan, announced to excited workers: "We're on the moon."

The officials said the spacecraft landed in the right position and was operating normally. The successful touchdown makes Firefly the first private company to land a spacecraft on the moon without crashing or having a major problem.

Private lunar lander Blue Ghost after touching down on the moon with a special delivery for NASA Sunday, March 2, 2025. (NASA/Firefly Aerospace via AP)

Only five countries can claim successful moon landings: Russia, the United States, China, India and Japan.

Blue Ghost is named after a rare kind of firefly found in the U.S. The four-legged lander is two meters tall and 3.5 meters wide. It launched in mid-January from Florida, carrying 10 experiments for the American space agency NASA.

In a statement, NASA said the 10 experiments are designed to operate on the surface of the moon for one lunar day, which is about 14 Earth days. The Associated Press reported that NASA paid \$101 million for the spacecraft and \$44 million for the equipment it carried.

It was the third mission under NASA's Commercial Lunar Payload Services (CLPS) program. The program aims to turn over the country's major space missions to private companies to reduce costs. Several other companies are part of the program.

Staff at the Mission Control outside Austin, Texas celebrating as lunar lander Blue Ghost touches down on the moon with a special delivery for NASA, Sunday, March 2, 2025. (NASA/Firefly Aerospace via AP)

Another lander, called Athena, is set to attempt a moon landing on March 6. That four-meter-tall spacecraft was built and operated by Houston-based Intuitive Machines. It will land on

another part of the moon, about 160 kilometers from the lunar south pole.

A third lander from private Japanese company ispace will attempt a moon landing in about three months. The lander, called Resilience, shared its rocket ride with Blue Ghost. But it took a longer path to the moon. The company is also attempting to land on the moon for the second time. Its first lander crashed in 2023.

NASA has said it aims to have two private landers launch to the moon each year, realizing some missions will fail. The space agency's top science officer is Nicky Fox. She told the AP the latest launches "open up a whole new way for us to get more science to space and to the moon."

This image provided by Firefly Aerospace on Feb. 26, 2025 shows Firefly Aerospace's Blue Ghost lunar lander orbiting the moon. (Firefly Aerospace via AP)

In the past, NASA's successful moon landings involving astronauts cost billions of dollars. But Firefly chief Jason Kim said the private companies now building and launching spacecraft have a limited budget and the spacecraft operate robotically.

Kim said everything went just as planned with Blue Ghost's landing. "We got some moon dust on our boots," he added.

I'm Bryan Lynn.

The Associated Press, Agence France-Presse and NASA reported on this story. Bryan Lynn adapted the reports for VOA Learning English.

Vocabulary

mission - n. (space exploration) การเดินทางของยานอวกาศไปยังเป้าหมายพร้อมกับภารกิจที่คาดว่าจะต้องดำเนินการ

boot - n. รองเท้าที่แข็งแรงซึ่งคลุมเท้าและส่วนหนึ่งของขา

Reading Comprehension Quiz

1. What is the main subject of the article's opening statement regarding the Blue Ghost mission?

- A) The difficulties of space travel.
- B) The successful landing of a private spacecraft on the moon.
- C) The costs associated with lunar missions.
- D) The naming of the Blue Ghost lander.

2. According to the article, what is the name of the lander and the company responsible for its construction?

- A) Athena, Intuitive Machines.
- B) Resilience, ispace.
- C) Blue Ghost, Firefly Aerospace.
- D) Lunar Explorer, NASA.

3. On which specific part of the moon's near side did the Blue Ghost spacecraft touch down?

- A) The lunar south pole.
- B) Mare Crisium.
- C) Florida.
- D) Austin, Texas.

4. Where did officials confirm the successful landing of the Blue Ghost, and who made the announcement to the workers?

- A) NASA headquarters; Nicky Fox.
- B) Firefly's Mission Control center near Austin, Texas; Will Coogan.
- C) Houston-based Intuitive Machines; the CEO.
- D) Cape Canaveral, Florida; a launch engineer.

5. What unique historical accomplishment did Firefly Aerospace achieve with the Blue Ghost mission?

- A) It was the first company to orbit the moon.
- B) It successfully launched the largest lander to date.
- C) It became the first private company to land a spacecraft on the moon without crashing or major problems.
- D) It was the first private company to carry experiments for NASA.

6. According to the article, how many countries have successfully landed on the moon?

- A) Three.
- B) Four.
- C) Five.
- D) Six.

7. From which U.S. state did the Blue Ghost lander launch?

- A) Texas.
- B) California.
- C) Florida.
- D) New York.

8. What was the primary purpose of the 10 items Blue Ghost carried for NASA?

- A) To collect moon rocks for return to Earth.
- B) To conduct various science experiments.
- C) To build a lunar base.
- D) To scout for potential human landing sites.

9. For how long are the experiments on Blue Ghost designed to operate on the lunar surface?

- A) Approximately 24 Earth hours.
- B) One lunar day, equivalent to about 14 Earth days.
- C) Several Earth months.
- D) Indefinitely, as long as power permits.

10. What was the total amount NASA reportedly paid for the Blue Ghost spacecraft and the equipment it carried?

- A) \$101 million.
- B) \$44 million.
- C) \$145 million.
- D) \$57 million.

11. What is the primary goal of NASA's Commercial Lunar Payload Services (CLPS) program?

- A) To launch more astronauts to the moon.
- B) To develop new rocket technology for space travel.
- C) To reduce costs by using private companies for major space missions.
- D) To discover new celestial bodies beyond the moon.

12. Which company is responsible for building and operating the Athena lander?

- A) Firefly Aerospace.
- B) ispace.
- C) Intuitive Machines.
- D) NASA.

13. Where is the Athena lander expected to touch down on the moon?

- A) Mare Crisium.
- B) Exactly at the lunar south pole.
- C) About 160 kilometers from the lunar south pole.

D) On the moon's far side.

14. What happened during the first moon landing attempt by the Japanese company ispace in 2023?

- A) It successfully landed but lost communication.
- B) It orbited the moon but did not attempt a landing.
- C) It crashed.
- D) It returned to Earth safely.

15. According to Nicky Fox, NASA's top science officer, what is the significance of the latest private lander launches?

- A) They will lead to manned missions being cheaper.
- B) They provide a new way to get more science to space and the moon.
- C) They prove that private companies are more reliable than government agencies.
- D) They indicate a shift away from lunar exploration.